

What is a Deployment?

Simple Definition

A Deployment is a Kubernetes object that manages Pods and ensures the desired number of Pods are always running.

Beginner Definition

A Deployment automatically creates, updates, and manages Pods.

Why Do We Need a Deployment?

Imagine you create **one Pod** directly.

Pod

If the Pod crashes or is deleted:

✗ It is gone.

Kubernetes will **not recreate it** automatically.

Now create a **Deployment**.

Deployment

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ReplicaSet

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Pod

If the Pod crashes:

Pod ✗ Crashed



Deployment notices



Creates a New Pod 

That's why Deployments are used in real projects.

Real-Life Example

Imagine a school.

-  Principal = Deployment
-  Teacher = Pod

The principal wants **3 teachers**.

If one teacher leaves:

Teacher 1 

Teacher 2 

Teacher 3 



Principal hires another teacher.

Teacher 1 

Teacher 2 

Teacher 3 

Similarly,

Deployment always maintains the required number of Pods.

Responsibilities of a Deployment

-  Creates Pods.
 -  Manages Pods.
 -  Updates Pods.
 -  Replaces failed Pods.
 -  Scales Pods up or down.
 -  Performs rolling updates.
 -  Supports rollback.
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Kubernetes Hierarchy

Deployment

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ReplicaSet

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Pod

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Container

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Application

Deployment vs Pod

Pod	Deployment
Runs an application	Manages Pods
Can fail and disappear	Recreates Pods automatically
No automatic updates	Supports rolling updates
No scaling	Supports scaling
Mostly used for testing	Used in production

Common Deployment Commands

1. View Deployments

`kubectl get deployments`

or

`kubectl get deploy`

2. Create a Deployment

`kubectl create deployment nginx-deployment --image=nginx`

Creates a Deployment using the **nginx** image.

3. Describe a Deployment

`kubectl describe deployment nginx-deployment`

Shows:

- Deployment details
- Events
- Replicas

- Strategy
 - Labels
-

4. Scale a Deployment

Increase Pods:

```
kubectl scale deployment nginx-deployment --replicas=5
```

Now Kubernetes creates **5 Pods**.

5. Check Pods Created by Deployment

```
kubectl get pods
```

6. Delete a Deployment

```
kubectl delete deployment nginx-deployment
```

Deletes the Deployment and its Pods.

7. View Deployment in YAML

```
kubectl get deployment nginx-deployment -o yaml
```

8. Update the Image

```
kubectl set image deployment/nginx-deployment nginx=nginx:latest
```

Performs a **Rolling Update**.

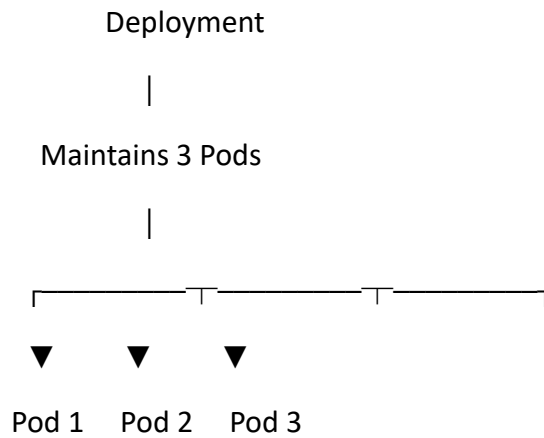
9. Check Rollout Status

```
kubectl rollout status deployment nginx-deployment
```

10. Roll Back to Previous Version

kubectl rollout undo deployment nginx-deployment

Easy Diagram



If **Pod 2** crashes:

Deployment

Pod 1 

Pod 2 

Pod 3 

↓

Deployment creates

Pod 2 

Important KCNA Points ★

- 📁 Deployment manages Pods.
 - 🔄 Automatically recreates failed Pods.
 - 📈 Supports scaling (increase/decrease Pods).
 - 🔄 Supports rolling updates.
 - ⏮ Supports rollback to a previous version.
 - 🏭 Used for production applications.
 - 📄 Deployment creates and manages ReplicaSets, which in turn manage Pods.
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Quick Revision Table

Component Easy Meaning

Deployment Manages Pods

ReplicaSet Maintains the required number of Pods

Pod Smallest deployable unit

Container Runs the application

Easy Memory Trick ★

Deployment

|



ReplicaSet

|



Pods

|



Containers



Application

One-Line Formula

Deployment → ReplicaSet → Pod → Container → Application

Remember

-  **Deployment = Manager**
-  **ReplicaSet = Keeps the correct number of Pods**
-  **Pod = Holds the container**
-  **Container = Runs the application**

This is the flow you'll encounter most often in Kubernetes deployments.

Many beginners confuse **Deployment** and **ReplicaSet**. The easiest way to remember is:

Deployment is the Manager, ReplicaSet is the Supervisor.

Deployment vs ReplicaSet

Deployment	ReplicaSet
Manages ReplicaSets	Manages Pods
Supports Rolling Updates	Does NOT support Rolling Updates
Supports Rollback	Does NOT support Rollback
Creates ReplicaSets automatically	Creates and maintains Pods
Used in Production	Usually managed by Deployment

Easy Real-Life Example 🏠

Imagine a school.

👤 Deployment = Principal

- Decides there should be **5 teachers**.
 - Can replace old teachers with new ones.
 - Can promote or update teachers.
 - Supervises everything.
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👩 ReplicaSet = Head Teacher

- Makes sure **5 teachers** are always present.
- If one teacher is absent, hires another.

It **doesn't decide policies**, it only maintains the required number.

Kubernetes Flow

Deployment

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Creates



ReplicaSet

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Creates & Maintains



Pods

|

Runs



Containers

Example

Suppose you want **3 Pods**.

You create a Deployment.

```
kubectl create deployment nginx --image=nginx --replicas=3
```

Kubernetes creates:

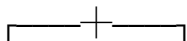
Deployment

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ReplicaSet

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Pod1 Pod2 Pod3

Now suppose **Pod2 crashes**.

Deployment

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ReplicaSet

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Pod1 

Pod2 

Pod3 

ReplicaSet notices that only **2 Pods** are running.

It immediately creates another Pod.

Deployment

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ReplicaSet

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Pod1 

Pod2  (New)

Pod3 

What if You Update the Application?

Suppose your application changes from:

nginx:1.24

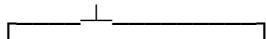
to

nginx:1.25

The **Deployment** creates a **new ReplicaSet**.

Deployment

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Old ReplicaSet New ReplicaSet

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Old Pods New Pods

After the update succeeds:

- Old ReplicaSet is scaled down.
- New ReplicaSet becomes active.

This is called a **Rolling Update**.

Responsibilities

Deployment

- Creates ReplicaSets.
- Performs Rolling Updates.
- Performs Rollbacks.
- Scales applications.
- Manages application versions.

ReplicaSet

- Creates Pods.
- Maintains the required number of Pods.
- Recreates Pods if they fail.

Easy Memory Trick ★

Deployment



"Boss"

ReplicaSet



"Pod Manager"

Pod



Application

Interview Answer (1 Line)

Deployment

Manages ReplicaSets and provides updates, rollbacks, and scaling.

ReplicaSet

Ensures the required number of identical Pods are always running.

Quick Revision Table

Feature	Deployment	ReplicaSet
Creates Pods	✗ (Indirectly)	✓
Creates ReplicaSet	✓	✗
Maintains Pod Count	✗	✓
Rolling Updates	✓	✗
Rollback	✓	✗
Production Use	✓ Yes	✗ Usually managed by Deployment

★ Memory Formula (Most Important for KCNA)

Deployment

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ReplicaSet

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Pods

|



Containers



Application

Remember this sentence:

Deployment manages ReplicaSets, ReplicaSets manage Pods, and Pods run Containers.